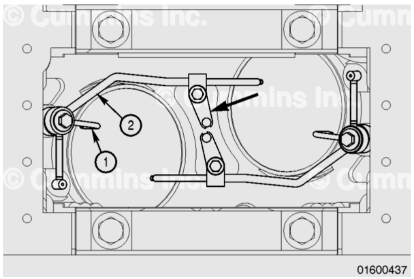
Trainings

General Information

QSK45 and QSK60 engines have a dual piston cooling nozzles configuration containing a short (1) and a long (2) nozzle. The nozzles mount on a pad between the cylinders and at an outboard location, as shown. The nozzle mounts on the pad between the cylinders with an M10 capscrew and at the outboard location with an M14 banjo capscrew.



There are two types of dual piston cooling nozzle available for QSK45 and QSK60 engines. These are referred to as thick based (old) and thin based (new) piston cooling nozzles.

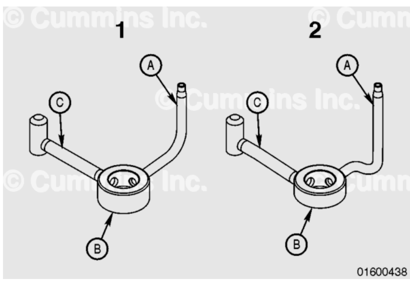
Thin based piston cooling nozzles were released for use with the straight split connecting rod. Refer to Procedure 001-014 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-014.html)

All QSK45 and QSK60 engines built after engine serial number (ESN) 33172060 have the thin based (new) nozzles.

Short Nozzle:

The differences between the thick based (1) and thin based (2) piston cooling nozzles are shown:

A. Nozzle tube shape
B. Banjo connection thickness
C. Positioning arm thickness and shape.



The thick based banjo connection (1B) is 16-mm [0.63-in] thick and the thin based banjo connection (2B) is 10-mm [0.39-in] thick. The thick based positioning arm is straight while the thin based positioning arm is slightly bent.

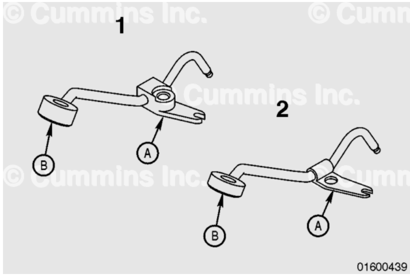
Long Nozzle:

The differences in the thick based (1) and thin based (2) piston cooling nozzles are shown:

A. Clamp thickness and shape
B. Banjo connection thickness.

Note : The thick based banjo connection (1) is 17.5-mm [0.69-in] thick and the thin based banjo connection (2) is 13.5-mm [0.53-in] thick.

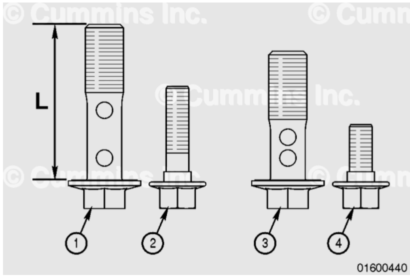
Note : The thick based nozzles can not be used on engines with straight split connecting rods.



Due to the difference in height of the clamp and banjo connection between the two sets of nozzles, the mounting capscrews are different in length. The correct capscrews **must** be used during installation. The capscrew and length for each nozzle are:

	Description	mm [in]
1	Banjo capscrew, thick based	57 [2.24]
2	Capscrew, thick based	35 [1.38]
3	Banjo capscrew, thin based	45 [1.77]
4	Capscrew, thin based	20 [0.79]

The thin based capscrew (4) **must** have Loctite™ on it. New capscrews come with Loctite™ applied. If old capscrews are used, Loctite™ 272 **must** be applied to the threads.



Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

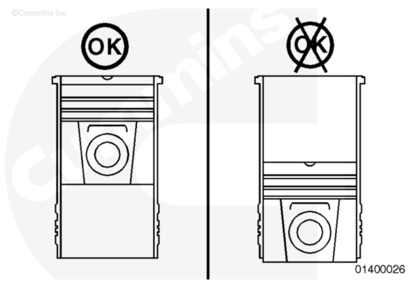
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Remove the oil pump suction and transfer tubes. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Remove the oil pump. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)
- Remove and discard the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-089-tr.html)

Remove

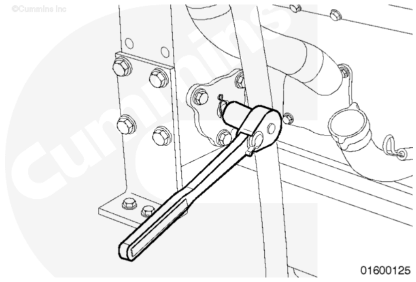
⚠ CAUTION ⚠

Do not attempt to remove the piston cooling nozzle when the piston is at bottom dead center. The nozzle will be damaged and can result in piston failure.



Rotate the engine, using the barring device, to position the pistons for best access to the piston cooling nozzles.

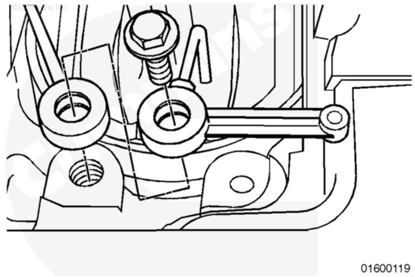
The left-bank crankshaft journal for the cylinder being worked on **must** be approximately 20 degrees before top dead center; the right-bank **must** be approximately 20 degrees after top dead center.



⚠ CAUTION ⚠

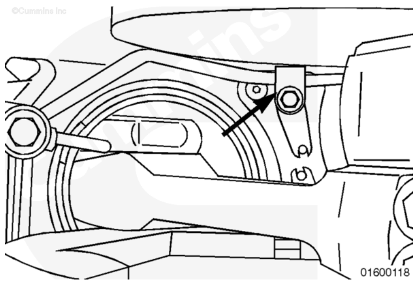
Handle the nozzles with care. Damage to the nozzles can result in piston failure.

The piston cooling nozzle is a dual-nozzle configuration that mounts on a pad between the cylinders and at an outboard location.



Remove the capscrew that mounts the long nozzle clip to the block at the center of the Vee.

Use a 13-mm [0.51-in] socket, 12-inch extension, and a long handled ratchet.

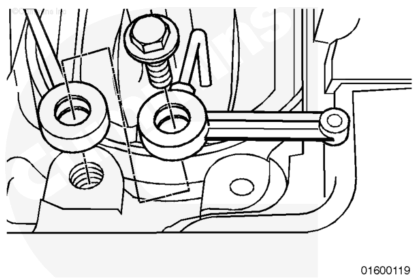


Remove the banjo capscrew that connects both nozzles on the outboard side of the cylinder block.

Remove both sections of the dual-piston cooling nozzle.

The cross bank nozzle is supported by a cap on the locating dowel. Do **not** pry down. Remove with caution.

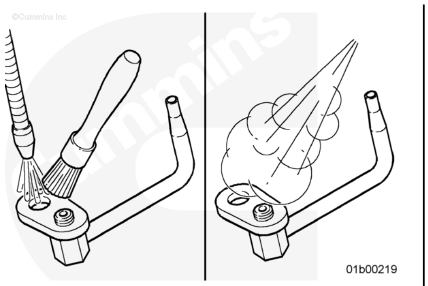
Piston cooling nozzles can easily be damaged and **must** be treated with care. Do **not** reuse piston cooling nozzles that have been bent or mutilated. Use of damaged piston cooling nozzles can result in severe engine damage.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

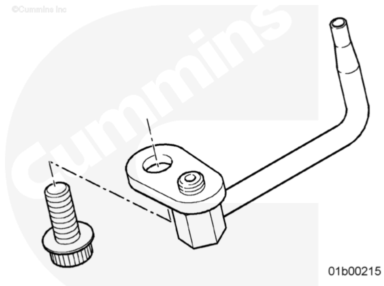
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Use solvent to clean the piston cooling nozzles. Dry with compressed air.

Blow out the oil passages with compressed air.

⚠ CAUTION ⚠

Any damage to the piston cooling nozzle can result in major engine damage.



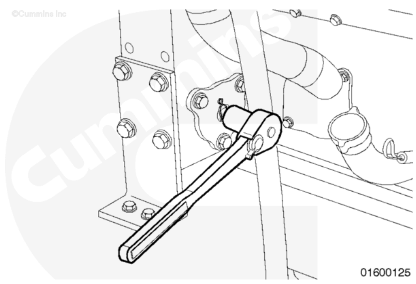
Check the nozzle for cracks, bends, or other damage. Check the spray hole for burrs.

The piston cooling nozzle **must** be replaced if it is damaged.

Install

⚠ CAUTION ⚠

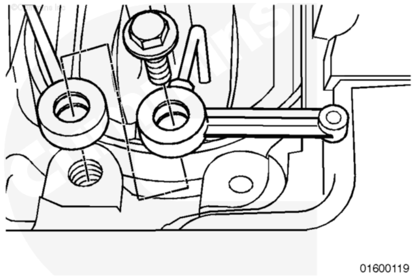
Do not attempt to install the piston cooling nozzle when the piston is at bottom dead center. The nozzle will be damaged and can result in piston failure.



Rotate the engine barring device to position the pistons for best access to the piston cooling nozzle mounting pad.

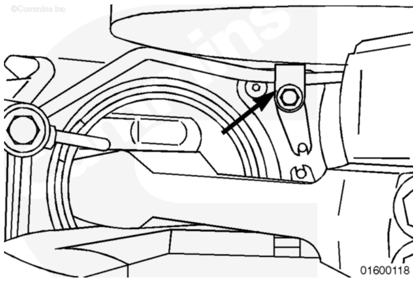
Gently install both sections of the dual-piston cooling nozzle, making sure the locating pin in the short section is properly placed into the cylinder block, and the longer section locating tab slides over the locating pin and is properly routed toward the adjacent cylinder.

Align the banjo connection of both piston cooling nozzle sections, and install the capscrew through both sections.



Install the capscrew that mounts the long nozzle clip to the cylinder block.

Tighten the capscrews.



Torque Value:
M14 Banjo Capscrew

1. 70 n•m [52 ft-lb]

Torque Value:
M10 Capscrew

1. 45 n•m [33 ft-lb]

The torque values for the thin and thick based nozzles are the same. When installing the thin based nozzles, the center capscrew containing Loctite™ **must** be run down and torqued at the same time. This prevents the Loctite® from setting before the final torque is applied.

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the lubricating oil pump. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)
- Install the lubricating oil pump suction and transfer tubes. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)

- Install the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Fill the engine with oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Operate the engine until the coolant temperature reaches 70°C [160°F].
- Check the engine for oil leaks.

Last Modified: 04-May-2015